

Multi PCSM

UL

Easy maintenance.

Integrated MV solution in the same enclosure.

Up to 4 independent DC inputs.

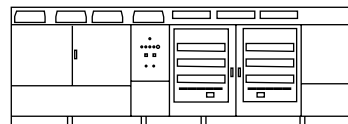
Advanced grid support.

Compatible with all battery technologies.



Freemaq Multi PCSM

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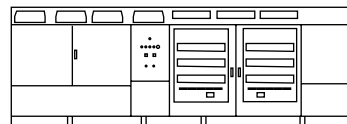
REFERENCES		FP4010M2	FP4010M4	FP4105M2	FP4105M4
AC	AC Output Power (kVA/kW) @40°C ^[5]	4010		4105	
	AC Output Power (kVA/kW) @50°C ^[5]	3720		3810	
	Operating Grid Voltage (kV)	34.5 kV ±10%			
DC	DC Voltage Range ^[6]	891 V - 1500 V		913 V - 1500 V	
	Max. DC Continuous Current per Input (A)	2295	1148	2295	1148
	Number of Separate DC Inputs	2	4	2	4
EFFICIENCY	Efficiency (Max) (η)	97.91% including MV transformer		97.93% including MV transformer	
	CEC (η)	97.49% including MV transformer		97.50% including MV transformer	
COMMON FEATURES					
AC	Operating Grid Frequency (Hz)	60 Hz			
	Current Harmonic Distortion (THDi)	< 3% per IEEE519			
	Power Factor (CosPhi) ^[1]	0.5 leading ... 0.5 lagging			
	Reactive Power Compensation	Four quadrant operation			
	Overload Capability ^[7]	166% - 100 ms / 150% - 5 s / 120% - 8 s / 110% - 15 s			
DC	Maximum DC Voltage	1500 V			
	DC Voltage Ripple	< 3%			
	Max. DC Short Circuit Current per Input (kA)	500 kA with a time constant of 1 ms			
	Battery Technology	All type of batteries (BMS required)			
CABINET	Dimensions [WxDxH] (ft)	21.3 x 6.5 x 7.5			
	Dimensions [WxDxH] (m)	6.5 x 2.0 x 2.3			
	Weight (lbs)	30865			
	Weight (kg)	14000			
	Type of Ventilation	Forced air cooling			
ENVIRONMENT	Degree of Protection	NEMA 3R			
	Operating Temperature Range ^[2]	From -40 °C to +60 °C, >40 °C power derating			
	Operating Relative Humidity Range	From 4% to 100%			
	Storage Temperature Range	From -40 °C to +60 °C			
	Max. Altitude (above sea level)	2000 m			
CONTROL	Communication Protocol	Modbus TCP			
	Power Plant Controller	Optional. Third party SCADA systems supported.			
	Keyed ON/OFF Switch	Standard			
PROTECTIONS	Ground Fault Protection	Insulation monitoring device			
	Humidity Control	Active heating			
	General AC Protection & Disconn.	38 kV MV switchgear (V)			
	General DC Protection & Disconn. ^[3]	High-speed fuses, Motorized DC disconnect switches			
	Overvoltage Protection	Type II for AC and Type I-II for DC			
STANDARDS	Safety	UL 1741 / CSA 22.2 No.1071-16			
	Installation	NEC 2023			
	Utility Interconnect ^[4]	UL 1741 SA & SB / RULE 21 / RULE 14H / IEEE 15471:2020			

NOTES

- [1] Consult P-Q charts available: $Q(kVA) = \sqrt{(S(kVA))^2 - P(kW)^2}$.
 [2] Optional available for temperatures below -25 °C.
 [3] Battery short circuit disconnection must be done on the battery side.
 [4] Consult Power Electronics for other applicable standards / grid codes.
 [5] Values at 1.00-Vac nom and CosPhi=1. Consult Power Electronics for derating curves.
 [6] Consult Power Electronics for derating curves. In the event of overvoltage in the grid, the minimum DC voltage will vary proportionally with the AC voltage.
 [7] Available from January 2027.

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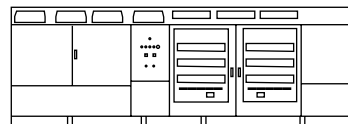
REFERENCES		FP4200M2	FP4200M4	FP4201M2	FP4201M4
AC	AC Output Power (kVA/kW) @40°C ^[5]	4200			
	AC Output Power (kVA/kW) @50°C ^[5]	3900			
	Operating Grid Voltage (kV)	34.5 kV ±10%		13.8 kV ±10%	
DC	DC Voltage Range ^[6]	934 V - 1500 V			
	Max. DC Continuous Current per Input (A)	2295	1148	2295	1148
	Number of Separate DC Inputs	2	4	2	4
EFFICIENCY	Efficiency (Max) (η)	98.00% including MV transformer			
	CEC (η)	97.53% including MV transformer			
COMMON FEATURES					
AC	Operating Grid Frequency (Hz)	60 Hz			
	Current Harmonic Distortion (THDi)	< 3% per IEEE519			
	Power Factor (CosPhi) ^[1]	0.5 leading ... 0.5 lagging			
	Reactive Power Compensation	Four quadrant operation			
	Overload Capability ^[7]	166% - 100 ms / 150% - 5 s / 120% - 8 s / 110% - 15 s			
DC	Maximum DC Voltage	1500 V			
	DC Voltage Ripple	< 3%			
	Max. DC Short Circuit Current per Input (kA)	500 kA with a time constant of 1 ms			
	Battery Technology	All type of batteries (BMS required)			
CABINET	Dimensions [WxDxH] (ft)	21.3 x 6.5 x 7.5			
	Dimensions [WxDxH] (m)	6.5 x 2.0 x 2.3			
	Weight (lbs)	30865			
	Weight (kg)	14000			
	Type of Ventilation	Forced air cooling			
ENVIRONMENT	Degree of Protection	NEMA 3R			
	Operating Temperature Range ^[2]	From -40 °C to +60 °C, >40 °C power derating			
	Operating Relative Humidity Range	From 4% to 100%			
	Storage Temperature Range	From -40 °C to +60 °C			
	Max. Altitude (above sea level)	2000 m			
CONTROL	Communication Protocol	Modbus TCP			
	Power Plant Controller	Optional. Third party SCADA systems supported.			
	Keyed ON/OFF Switch	Standard			
PROTECTIONS	Ground Fault Protection	Insulation monitoring device			
	Humidity Control	Active heating			
	General AC Protection & Disconn.	38 kV MV switchgear (V)		15.5 kV SF6 Free MV switchgear (V)	
	General DC Protection & Disconn. ^[3]	High-speed fuses, Motorized DC disconnect switches			
	Overvoltage Protection	Type II for AC and Type I-II for DC			
STANDARDS	Safety	UL 1741 / CSA 22.2 No.1071-16			
	Installation	NEC 2023			
	Utility Interconnect ^[4]	UL 1741 SA & SB / RULE 21 / RULE 14H / IEEE 15471:2020			

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- [1] Consult P-Q charts available: $Q(kVA) = \sqrt{(S(kVA))^2 - P(kW)^2}$.
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 [7] Available from January 2027.

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REFERENCES		FP4204M2	FP4204M4
AC	AC Output Power (kVA/kW) @40°C ^[5]	4200	
	AC Output Power (kVA/kW) @50°C ^[5]	3900	
	Operating Grid Voltage (kV)	12.47 kV ±10%	
DC	DC Voltage Range ^[6]	934 V - 1500 V	
	Max. DC Continuous Current per Input (A)	2295	1148
	Number of Separate DC Inputs	2	4
EFFICIENCY	Efficiency (Max) (η)	98.00% including MV transformer	
	CEC (η)	97.53% including MV transformer	
COMMON FEATURES			
AC	Operating Grid Frequency (Hz)	60 Hz	
	Current Harmonic Distortion (THDi)	< 3% per IEEE519	
	Power Factor (CosPhi) ^[1]	0.5 leading ... 0.5 lagging	
	Reactive Power Compensation	Four quadrant operation	
	Overload Capability ^[7]	166% - 100 ms / 150% - 5 s / 120% - 8 s / 110% - 15 s	
DC	Maximum DC Voltage	1500 V	
	DC Voltage Ripple	< 3%	
	Max. DC Short Circuit Current per Input (kA)	500 kA with a time constant of 1 ms	
	Battery Technology	All type of batteries (BMS required)	
CABINET	Dimensions [WxDxH] (ft)	21.3 x 6.5 x 7.5	
	Dimensions [WxDxH] (m)	6.5 x 2.0 x 2.3	
	Weight (lbs)	30865	
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	Type of Ventilation	Forced air cooling	
ENVIRONMENT	Degree of Protection	NEMA 3R	
	Operating Temperature Range ^[2]	From -40 °C to +60 °C, >40 °C power derating	
	Operating Relative Humidity Range	From 4% to 100%	
	Storage Temperature Range	From -40 °C to +60 °C	
	Max. Altitude (above sea level)	2000 m	
CONTROL	Communication Protocol	Modbus TCP	
	Power Plant Controller	Optional. Third party SCADA systems supported.	
	Keyed ON/OFF Switch	Standard	
PROTECTIONS	Ground Fault Protection	Insulation monitoring device	
	Humidity Control	Active heating	
	General AC Protection & Disconn.	15.5 kV SF6 Free MV switchgear (V)	
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	Overvoltage Protection	Type II for AC and Type I+II for DC	
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